Chapter 4: Continuity

Section: Limits of Functions

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

Motivation

From Sequences to Functions

- Chapter 3: limits of sequences — $\lim_{n\to\infty} p_n = p$

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- Chapter 3: limits of sequences — $\lim_{n \to \infty} p_n = p$
- Now: limits of functions — $\lim_{x \to p} f(x) = q$

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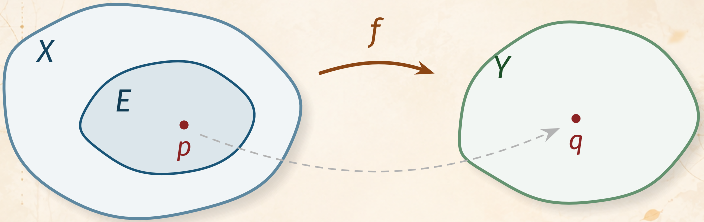
- Now: limits of functions — $\lim_{x \to p} f(x) = q$
- Bridge: Theorem 4.2 reduces function limits to sequence limits

The General Setting



Setting: f maps $E \subset X$ into Y, where X, Y are metric spaces.

Why so general? Real, complex, and vector-valued functions are all special cases — the general proofs are *no harder*.

The Definition of a Limit

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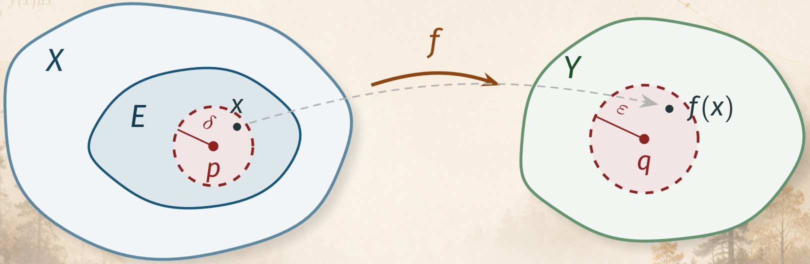
if there is a point $q \in Y$ with the following property: for every $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$d_Y(f(x), q) < \varepsilon \tag{2}$$

for all points $x \in E$ for which

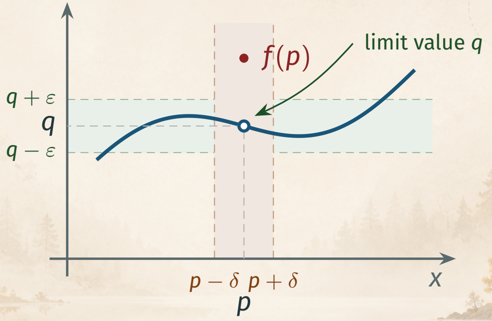
$$0 < d_X(x, p) < \delta. \tag{3}$$

The Picture in Metric Spaces



Reading the definition: every point $x \neq p$ inside the δ-ball (and in E) is sent by f into the ε-ball around q.

A Limit Where $f(p) \neq q$



p need not be in E; even if $p \in E$, possibly $f(p) \neq \lim_{x \to p} f(x)$.

Remarks on Definition 4.1

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- $p \in X$ must be a limit point of E, but p need not belong to E.
- Even if $p \in E$, possibly $f(p) \neq \lim_{x \to p} f(x)$: condition (3) excludes $x = p$.

Limits via Sequences

Theorem 4.2: Sequential Characterization

Theorem 4.2

Let X, Y, E, f, and p be as in Definition 4.1. Then

$$\lim_{x \to p} f(x) = q \tag{4}$$

if and only if

$$\lim_{n \to \infty} f(p_n) = q \tag{5}$$

for every sequence $\{p_n\}$ in E such that

$$p_n \neq p, \qquad \lim_{n \to \infty} p_n = p. \tag{6}$$

Proof Strategy: Theorem 4.2

Goal: Prove the equivalence (4) $\iff$ (5).

1. **($\Rightarrow$)** Assume (4). Chain the two definitions: $\varepsilon \rightsquigarrow \delta \rightsquigarrow N$.
2. **($\Leftarrow$)** Prove the contrapositive: if (4) fails, build a bad sequence with $\delta_n = 1/n$.

Technique: the second half is a classic sequence-construction.

Proof of Theorem 4.2 — $(\Rightarrow)$

Step 1: From ε to δ

Suppose (4) holds. Choose $\{p_n\}$ in E satisfying (6). Let $\varepsilon > 0$ be given. Then there exists $\delta > 0$ such that

$$d_Y(f(x), q) < \varepsilon \quad \text{if } x \in E \text{ and } 0 < d_X(x, p) < \delta.$$

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Step 2: From δ to N

Since $p_n \to p$ and $p_n \neq p$, there exists N such that

$$n > N \implies 0 < d_X(p_n, p) < \delta \implies d_Y(f(p_n), q) < \varepsilon.$$

Hence $f(p_n) \to q$, i.e., (5) holds. ■ (forward direction)

Proof of Theorem 4.2 — ($\Leftarrow$), by Contraposition

Step 1: Negate (4)

Suppose (4) is false. Then there exists some $\varepsilon > 0$ such that for every $\delta > 0$ there is a point $x \in E$ (depending on δ) with

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Step 2: Build the bad sequence

Take $\delta_n = \frac{1}{n}$ $(n = 1, 2, 3, \ldots)$: we obtain points $p_n \in E$ with $0 < d_X(p_n, p) < \frac{1}{n}$ but $d_Y(f(p_n), q) \geq \varepsilon$.

The sequence $\{p_n\}$ satisfies (6), yet (5) is false. $\blacksquare$

Corollary: Uniqueness of Limits

Corollary

If f has a limit at p, this limit is unique.

Why: Theorem 3.2(b) (limits of sequences are unique) $+$ Theorem 4.2.

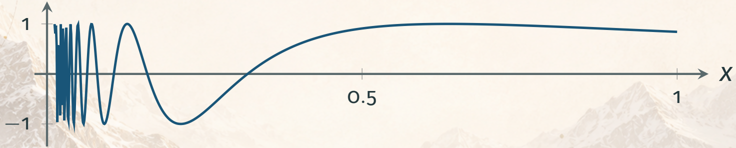
Recall: Theorem 3.2(b)

A sequence in a metric space converges to *at most one* point.

Example: A Limit That Fails to Exist

Example (the criterion in action)

Does $\lim_{x \to 0} \sin\left(\frac{1}{x}\right)$ exist? Here $E = (0, 1]$; note $0 \notin E$, but 0 *is* a limit point of E.



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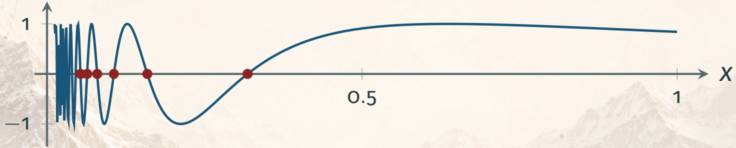

- Along $a_n = \frac{1}{n\pi} \to 0$: $f(a_n) = \sin(n\pi) = 0 \;\to\; 0$

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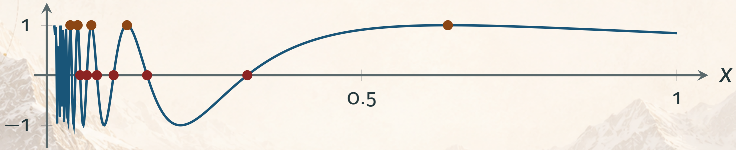

- Along $a_n = \frac{1}{n\pi} \to 0$: $f(a_n) = \sin(n\pi) = 0 \ \to \ 0$
- Along $b_n = \frac{2}{(4n+1)\pi} \to 0$: $f(b_n) = \sin\left(2n\pi + \frac{\pi}{2}\right) = 1 \ \to \ 1$

Two sequences, two different limits $\implies$

by Theorem 4.2, the limit does not exist.

The Algebra of Functions

Definition 4.3: Operations on Functions

Definition 4.3 (complex functions)

Let $f, g\colon E \to \mathbb{C}$. Define, for each $x \in E$:

$$(f+g)(x) = f(x) + g(x), \qquad (f-g)(x) = f(x) - g(x),$$

$$(fg)(x) = f(x)\,g(x), \qquad \left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} \text{ (only where } g(x) \neq 0\text{)}.$$

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Notation

- $f = c$: *constant* function, $f(x) = c$ for all $x \in E$.
- $f \geq g$ (real f, g): means $f(x) \geq g(x)$ for every $x \in E$.

Definition 4.3: Vector-Valued Functions

Definition 4.3 (vector-valued functions)

If $\mathbf{f}, \mathbf{g} \colon E \to \mathbb{R}^k$, define

$$(\mathbf{f} + \mathbf{g})(x) = \mathbf{f}(x) + \mathbf{g}(x), \qquad (\mathbf{f} \cdot \mathbf{g})(x) = \mathbf{f}(x) \cdot \mathbf{g}(x),$$

and for $\lambda \in \mathbb{R}$, $(\lambda \mathbf{f})(x) = \lambda \mathbf{f}(x)$.

Note: $\mathbf{f} \cdot \mathbf{g}$ is the inner product — a *scalar*-valued function.

Theorem 4.4: Limit Laws

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Suppose $E \subset X$, a metric space, p is a limit point of E, f and g are complex functions on E, and

$$\lim_{x \to p} f(x) = A, \qquad \lim_{x \to p} g(x) = B.$$

Then:

(a) $\lim_{x \to p} (f + g)(x) = A + B$;

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Then:

(a) $\lim_{x \to p} (f + g)(x) = A + B$;

(b) $\lim_{x \to p} (fg)(x) = AB$;

(c) $\lim_{x \to p} \left(\frac{f}{g}\right)(x) = \frac{A}{B}$, if $B \neq 0$.

Proof of Theorem 4.4

Proof

In view of Theorem 4.2, these assertions follow immediately from the analogous properties of sequences (Theorem 3.3). ■

The bridge in action:

$$\text{(4) function limit} \overset{\text{Thm 4.2}}{\longleftrightarrow} \text{(5) sequence limits} \overset{\text{Thm 3.3}}{\longleftrightarrow} \text{limit laws}$$

Remark: Vector-Valued Limit Laws

Remark

If $\mathbf{f}$ and $\mathbf{g}$ map E into $\mathbb{R}^k$, then (a) remains true, and (b) becomes:

(b') $\lim_{x \to p} (\mathbf{f} \cdot \mathbf{g})(x) = \mathbf{A} \cdot \mathbf{B}$.

(Compare Theorem 3.4.) There is no analogue of (c): we cannot divide by a vector.

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Key takeaways from this section:

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- Definition 4.1: $\lim_{x\to p} f(x) = q$ via ε–δ in metric spaces; p a limit point of E, value $f(p)$ *irrelevant*
- Theorem 4.2: function limits $\iff$ sequence limits; corollary: limits are *unique*
- Theorem 4.4: limit laws for $f+g$, fg, f/g (and $\mathbf{f}+\mathbf{g}$, $\mathbf{f}\cdot\mathbf{g}$)

Coming up next:

Continuous Functions — what happens when $\lim_{x\to p} f(x) = f(p)$.